THE GORENSTEIN QUOTIENT $\mathrm{QSym}/\operatorname{Ann}(R_s)$ FOR ALTERNATING RIBBONS: HILBERT FUNCTION (BOTH PARITIES) AND RING STRUCTURE (Q75)

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ring computations `proofs/scripts/q75_ring.py`.

Summary. For a ribbon $R_s \in \mathrm{NSym}$ let $G_s = \mathrm{QSym}_{\mathbb{Q}}/\operatorname{Ann}(R_s)$ be the graded Artinian Gorenstein algebra with Macaulay dual generator R_s. Its Hilbert function is the rank function of the cap product $\cap[X_s]$ on the toric Richardson variety X_s of Gonzales–Le–Lowen (arXiv:2609.30257). For alternating s we give closed forms for both parities of the length (Theorem 1), and show that $\mathrm{Sym} \to G_s$ is onto (Corollary 2). In the odd case $s = (2^k)$ we have $\mathrm{Hilb}(G_s) = \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_q$, which is the Poincaré polynomial of $\mathrm{Gr}(2, k+2)$. Nevertheless $G_{(2^k)}$ is *not* isomorphic to $H^*(\mathrm{Gr}(2, k+2))$ for $k = 3, 4, 5$ (§4).

Grades.

- proved: Theorem 1 and Corollary 2. This is an internal grade, after one internal cold read (PASS, 0 must-fix).
- computed ($k \le 5$): the ring-structure statements in §4.
- sketched: the complete-intersection claim for general k.
- Owner check (2026-11-12): not found on either the QSym side or the Sym side (§5).

What I want from you:

- a check of the Key Identity (§3);
- a pointer if you know of a source on inverse systems of ribbon or skew Schur functions.

1. Setting

$\langle\,,\rangle$ is the QSym–NSym pairing: F_α is dual to R_α, and M_α is dual to S^α. For $f \in \mathrm{QSym}$ define $f^\perp$ on NSym by $\langle g, f^\perp a\rangle = \langle fg, a\rangle$. A binary string s of length L encodes a composition of $d = L+1$; we call s *alternating* if it is $0101\cdots$ or $1010\cdots$. Put

$$r_i(s) = \operatorname{rank}\big(\mathrm{QSym}_i \times \mathrm{QSym}_{d-i} \to \mathbb{Q},\ (f,g) \mapsto \langle fg, R_s\rangle\big) = \dim(G_s)_i.$$

Complementation of descent sets is an algebra automorphism of QSym, so $r(s) = r(s^c)$. We may therefore take $s = (2^k)$ (odd $L = 2k-1$) or $s = (2^k, 1)$ (even $L = 2k$).

2. Results

Theorem 1. *Let s be alternating of length L.*

(a) If $L = 2k-1$ (so $d = 2k$), then $r_i = \min(\lfloor i/2\rfloor, \lfloor (d-i)/2\rfloor) + 1$, i.e. $\mathrm{Hilb}(G_s; q) = [\lfloor k/2\rfloor + 1]_{q^2}\,[2\lceil k/2\rceil + 1]_q = \begin{bmatrix} k+2 \\ 2 \end{bmatrix}_q$.

(b) If $L = 2k$ (so $d = 2k+1$), then $r_0 = r_d = 1$ and $r_i = \min(\lceil i/2\rceil, \lceil (d-i)/2\rceil)$ for $0 < i < d$.

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Corollary 2. *For alternating s the composite* $\mathrm{Sym} \hookrightarrow \mathrm{QSym} \twoheadrightarrow G_s$ *is onto. Hence the apolar algebra* $\mathrm{Sym}/\operatorname{Ann}_{\mathrm{Sym}}(\chi R_s)$ *under the Hall pairing has the same Hilbert function. For odd L, $\chi R_s = r_{(2^k)}$ is the ribbon Schur function with all rows of length 2.*

The formula in (b) is new. It replaces a fitted numerator from the scoping session and agrees with the brute-force table for all $L \le 22$. Formula (a) was first predicted blind at $L = 13$ and then matched for $L \le 20$.

3. The one idea

Let $\sigma(z) = \sum_n S_n z^n$, and let Ψ and Θ be its even and odd parts. Set $W = \Psi^{-1} = \sum_k (-1)^k z^{2k} R_{(2^k)}$. The coproduct of M_α is deconcatenation, so $M_\alpha^\perp$ acts on a product by lowering factors at strictly increasing positions. Even parts collapse because $W\Psi = 1$. This gives the **Key Identity**

$$M_\alpha^\perp W = (-1)^{\ell(\alpha)}\, z^{|\alpha|}\, (W\Theta)^{o(\alpha)}\, W, \qquad o(\alpha) = \#\{\text{odd parts of } \alpha\}.$$

Consequently $M_\alpha^\perp R_{(2^k)}$ depends only on $(\ell(\alpha), o(\alpha))$, which gives the upper bound $\lfloor i/2 \rfloor + 1$.

For the lower bound, pass to the commutative image with $p_{\text{even}} \mapsto 0$ except $p_2 \mapsto y$. The images become $e^{-yz^2/2}\operatorname{sech}(A)\tanh^o(A)$, and these are independent by an order and bidegree argument.

The even case runs the same way, using $W\Theta = \sum_k (-1)^k z^{2k+1} R_{(2^k,1)}$. Here $M_\alpha^\perp R_{(2^k,1)} = 0$ whenever α ends in an even part. The Key Identity was checked by machine on 1364 of 1364 (odd, $k \le 5$) and 10915 of 10915 (even) coefficient instances; the proof itself does not rely on these checks.

4. Ring structure: complete intersection, but not Grassmannian (computed)

All statements in this section are computed with exact rational arithmetic for $k \le 5$ (`q75_ring.py`, about 6 s), except where marked sketched.

Presentation. Take $s = (2^k)$. Using $p_n^\perp = n\,\partial/\partial p_n$, one sees that G_s is generated by $x = p_1$ (degree 1) and $y = p_2$ (degree 2). In general k this is sketched: on the derivative module, $p_{2n}^\perp = -(-p_2^\perp)^n$ and $p_{2n+1}^\perp = (-p_2^\perp)^n p_1^\perp$. A Gorenstein quotient of codimension two is a complete intersection (Serre). So $G_s = \mathbb{Q}[x,y]/(f_{k+1}, g_{k+2})$, which has the same presentation type as $H^*(\mathrm{Gr}(2,k+2)) = \mathbb{Q}[e_1,e_2]/(h_{k+1}, h_{k+2})$. The computed generators are:

k	generators of $I = \ker(\mathbb{Q}[x,y] \to G_{(2^k)})$
2	$x^3 - 5xy, \quad x^4 - 5y^2$
3	$x^4 - 14x^2y + 9y^2, \quad x^5 - \frac{61}{5}x^3y$
4	$x^5 - 30x^3y + 89xy^2, \quad x^6 - \frac{331}{14}x^4y + \frac{801}{14}y^3$
5	$x^6 - 55x^4y + 439x^2y^2 - 225y^3, \quad x^7 - \frac{4670}{89}x^5y + \frac{32321}{89}x^3y^2$

In these coordinates the Macaulay dual generator is $F = [z^{2k}]\, e^{Yz^2/2}\sec(Xz)$, where x acts as ∂_X and y acts as $2\partial_Y$.

Comparison with $H^*(\mathrm{Gr}(2,k+2))$. We look for a graded automorphism $x \mapsto ax$, $y \mapsto by + cx^2$ carrying I onto (h_{k+1}, h_{k+2}); the torus lets us normalise $a = 1$.

- $k = 2$: $G \cong H^*(\mathrm{Gr}(2,4);\mathbb{Q})$ via $y \mapsto (4e_2 - e_1^2)/5$. *This carries no information.* With Hilbert function $1,1,2,1,1$, every such complete intersection is isomorphic to $H^*(\mathrm{Gr}(2,4))$ over $\mathbb{C}$, because the moduli is trivial. Several non-alternating ribbons give algebras that are isomorphic to it over $\mathbb{C}$ but not over $\mathbb{Q}$.
- $k = 3$: two choices of (b,c) match in degree 4, namely $b = \pm 4\sqrt{2}/9$, $c = 7/9 \mp 2\sqrt{2}/3$. Both fail in degree 5: the relevant determinant is $-382/9 \mp 1342\sqrt{2}/45 \neq 0$. So $G_{(2,2,2)} \ncong H^*(\mathrm{Gr}(2,5))$ over $\mathbb{C}$.
- $k = 4$: no solution over $\mathbb{C}$.

- $k = 5$: the root sets of the degree-6 generators, written in $t = y/x^2$, are not affinely equivalent. So no isomorphism exists over $\mathbb{C}$.
- *Control.* For the rectangle $s_{(k,k)}$, the same code returns $H^*(\mathrm{Gr}(2, k+2))$ over $\mathbb{Q}$ for $k = 2, \ldots, 5$, via $y \mapsto p_2 = e_1^2 - 2e_2$. This reproduces the folklore fact $\mathrm{Sym}/\operatorname{Ann}(s_{(k^n)}) \cong H^*(\mathrm{Gr})$.

Conclusion. $G_{(2^k)}$ is a complete intersection with the Hilbert series and generator degrees of $H^*(\mathrm{Gr}(2, k+2))$. It is a different point of the same moduli space. The "fake rectangle" coincidence holds for Hilbert series only, not for rings.

Unlike H^* of a smooth projective toric variety, G_s needs a degree-2 generator. (Hunch, unchecked: this may reflect X_s not being rationally smooth.)

The even case $(2^k, 1)$ behaves differently. All p_{even} lie in Ann, and once $k \geq 5$ the element p_5 is a new generator. The embedding dimension is then at least 3, so a complete intersection is not automatic. This is sketched.

5. Owner check

Not found (2026-11-12). We searched both the QSym side and the Sym side, using the arXiv API, MathOverflow and web search; Semantic Scholar was rate-limited and is unsearched. The nearest results:

- Bergeron–Gagnon–Spink–Tewari 2604.24903: quotients of QSym, with no Poincaré duality. I would cite it.
- Nadeau–Spink–Tewari 2410.12643, Thm. 10.2 and Cor. 10.3: the Hall pairing with a ribbon skew Schur function equals a degree on a Grassmannian Richardson variety. There is no annihilator or Hilbert-series content.
- Almousa–Lu 2601.13324: single skewing operators lifted to modules. They do not consider spans or annihilators.
- Macias Marques–McDaniel–Seceleanu–Watanabe 2208.05653: codimension-two Artinian Gorenstein algebras have the SLP, in the standard grading only.

The rectangle statement $\mathrm{Sym}/\operatorname{Ann}(s_{(k^n)}) \cong H^*(\mathrm{Gr})$ is folklore; I did not locate an explicit citation for it.

6. What I am least sure of

- **Single internal check.** Theorem 1 has had one internal cold read and no external check. The Key Identity is the load-bearing step.
- **Only finitely many k are excluded.** The non-isomorphism is computed for $k = 3, 4, 5$ only. I have no conceptual reason why it should persist for all $k \geq 3$, although nothing suggests otherwise.
- **General k.** The complete-intersection presentation for general k is sketched: two-generation from the derivative-module rules, plus Serre.
- **Non-alternating s.** Whether $r_2 = 1$ exactly for s constant or even alternating is computed for $L \leq 14$ and otherwise open, as is the general-s formula.